

Ting-Ting GAO

📍 Boston, US | 🎓 [Google Scholar](#)
✉ ttinggao314@gmail.com
☎ +1 8573565890

EDUCATION

TONGJI UNIVERSITY

September 2014 - June 2018

B.Sc. in Applied Physics, *Cum Laude* (Tongji University)

TONGJI UNIVERSITY

September 2018 - March 2024

Ph.D. in Physics, *Cum Laude* (Shanghai)

Advisor: Prof. Gang Yan

- 2022 National Scholarship; 2022 Excellent Student Leader.

RESEARCH INTERESTS

- Network Science; Nonlinear Dynamics; AI for Science; Physical Network.

PUBLICATIONS

- **T.-T. Gao** & G. Yan*, Autonomous inference of complex network dynamics from incomplete and noisy data, **Nature Computational Science**, 2(3), 160-168, (2022). [Link](#)
- **T.-T. Gao** & G. Yan*, Data-driven inference of complex system dynamics: A mini-review, **Euro-physics Letters**, (2023). [Link](#)
- **T.-T. Gao**, B. Barzel, & G. Yan*, Learning interpretable dynamics of stochastic complex systems from experimental data, **Nature Communications**, 15(1), 6029, (2024). [Link](#)
- T. Ma, W. Cui, & **T.-T. Gao** et al, Data-based autonomously discovering method for nonlinear aerodynamic force of quasi-flat plate, **Physics of Fluids**, (2023). [Link](#)
- Z. Liu, X. Zhang, X. Ru, & **T.-T. Gao** et al, Early predictor for the onset of critical transitions in networked dynamical systems, **Physical Review X**, 14(3), 031009 (2024). [Link](#)
- X.-J. Zhang, JM Moore, **T.-T. Gao** et al, *Brain-inspired wiring economics for artificial neural networks*, **PNAS Nexus**, 4(1), pgae580 (2025). [Link](#)
- Y.-T. Sun, **T.-T. Gao*** et al, *Simultaneous Inference of Network Structure and Dynamics: How Much Data is Needed?*, in preparation.
- T. Ma, **T.-T. Gao** et al, *Encoding cumulation to learn perturbative nonlinear oscillatory dynamics*, submitted.

PRESENTATIONS

- **Invited talk**, at Swarma Club, "Interpretable dynamical inference and critical point analysis in complex systems", October 31, 2024.
- **Regular talk**, at NetSci 2023, "Inferring the stochastic dynamics of complex networks via message-passing mechanism", July 12 2023.
- **Regular talk**, at NetSci 2022, "Autonomous inference of complex network dynamics from incomplete and noisy data", July 27 2022.
- **Invited guest talk**, at Technical University of Dresden, "Discovering the complex network dynamics from incomplete and noisy data", May 3 2022.
- **Invited report**, at Swarma Club, "Discovering the complex network dynamics from incomplete and noisy data", April 16 2022.

EXPERIENCES

- 2018, **Teaching assistant**, General Physical Experiment, Tongji University
- 2019 - 2021, **Teaching assistant**, Network Science, Tongji University
- 2021 - 2022, **Teaching assistant**, AI and Physics, Tongji University
- 2020 - Present, **Reviewer** for Physical Review series, Chaos, IEEE Transactions on Network Science and Engineering, Nature Communications, etc.
- April 2024 - Present, **Postdoctoral researcher** under the supervision of Professor A.-L. Barabási at the CCNR Lab, Northeastern University.